DP Concepts & Questions

Video 38

भाषण
(Motivation)

Work hard in silence.
Build quietly, rise steadily.
When everyone finally
notice you, it will already
be too late to doubt
you...

MIK...

cswithMIK → Twitter
Facebook, Instagram → codestorywithMIK
whatsapp → codestorywithMIK

- 1-D based DP
- Grid based DP
- String based DP
- Knapsack Series
- Digit DP
- Game Strategy

We'll do :-

(·) RECURSION + MEMOIZATION (Top Down)

(·) Bottom Up.

(·) Time & Space

Knapsack Series

(video -3)

Variants

- Fractional Knapsack
- 0/1 Knapsack
- Unbounded Knapsack

W = 4

weight = {1, 4, 5}

profit = [3, 1, 2}

0-1 Knapsack

What if we are not allowed to take fractions of items ???

What if we pick least weights first ??

$W = 5$

weight = {1, 2, 3}

Profit = {1, 2, 4}

1 + 2

Profit = 1 + 2 = 3

4 + 2 = 6

What if we pick most profitable first??

$W = 10$ 40

weight = {9, 6, 4}

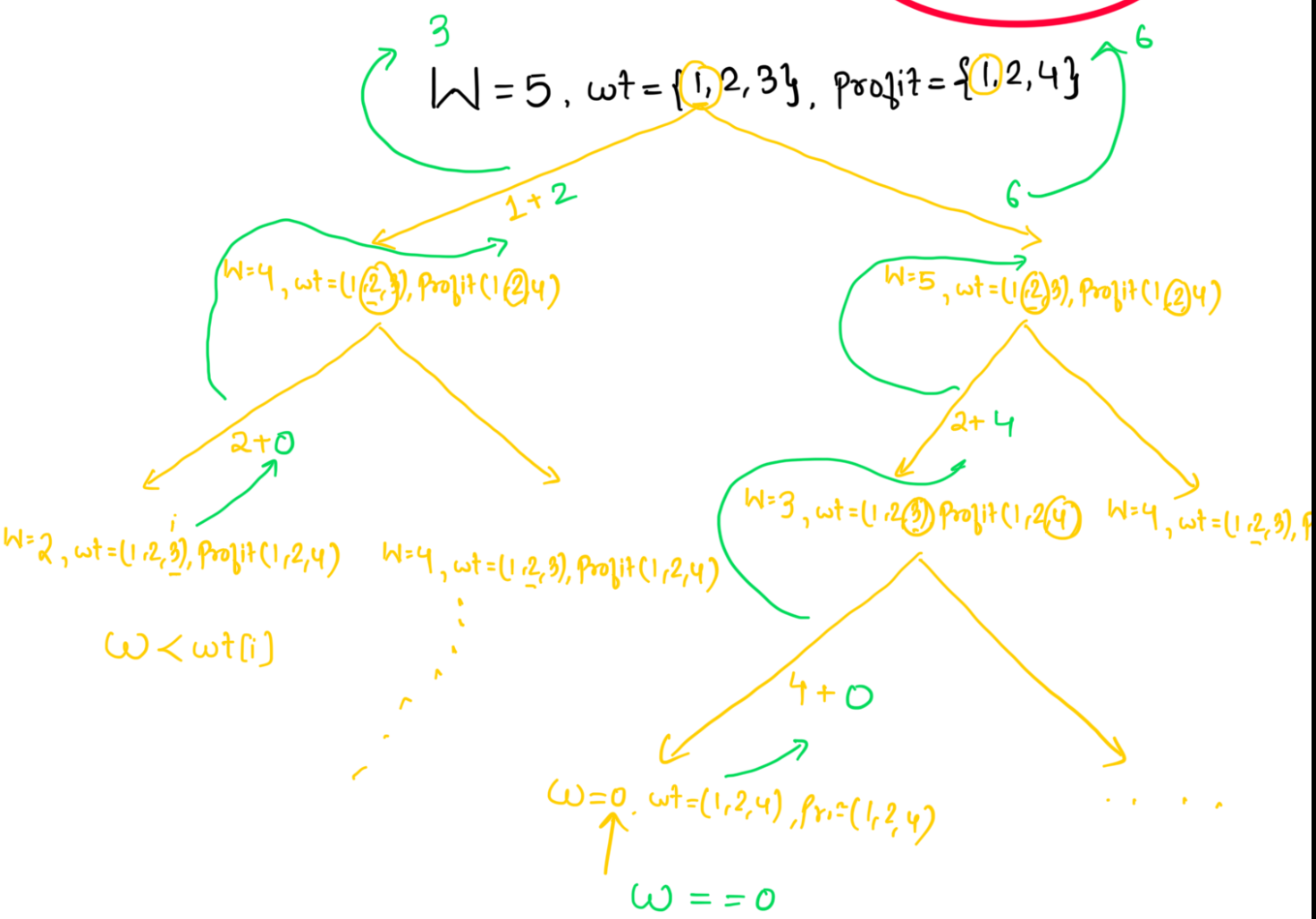
Profit = {100, 60, 50}

Profit = 100

optimal → 60 + 50 = 110

So, we will have to explore all

possibilities. (Recursively).



ω = 5 , wt = {1, 2, 3} , pr = {1, 2, 4}

W = 5 2+3 = 5 , 2+4 = 6

Recursion Story to Code

But, before that, a very important lesson to remember

```
Solve(i=0, wt, val, W);
```

length = n

n

```
Solve(i, wt, val, W) {
    if (i >= n || W == 0)
        ret 0;
    if (wt[i] <= W)
        take = val[i] + Solve(i+1, wt, val, W-wt[i]);
    Solve(i+1, wt, val, W);
```

i = index

wt

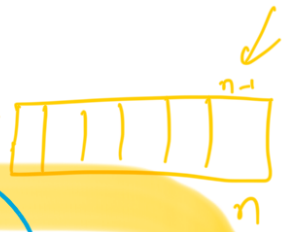
val

W

skip = solve(

}

n elements/items wt



n items
wt
val
W

```
Solve (n, wt, val, W) {
    if (n == 0 || W == 0)
        return 0;

    if (wt[n-1] <= W)
        take = val[n-1] + solve(n-1, wt, val, W-wt[n-1]);

    skip = solve(n-1, wt, val, W);
}
```

Bottom UP

Bottom UP...